WORKSHEET

Chapter 2: Modulus and argument

How far that little candle throws his beams …

13 10   13 3 5 2 9 13   1   6 10 10 12

12 9 9 12   5 2   1   11 9 1 8 7

11 10 8 4 12

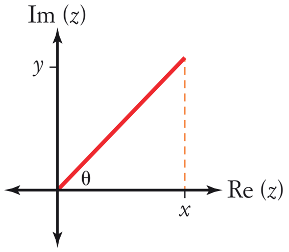
William Shakespeare

Rules

The modulus, $|z|$, of $z = x + yi$ is calculated using:

$$|z| = \sqrt{x^2 + y^2}$$

The argument, $\arg(z)$, is the angle θ from the positive real axis where $-\pi < \theta \leq \pi$.



Questions

Calculate the modulus of the following complex numbers:

1 $z = 5 - 12i$

2 $z = 3 + 4i$

3 $z = 7 - \sqrt{3}\,i$

4 $z = 6 + 2i$

Given $a = 4 + 3i$, $b = 5 - 2i$ and $c = 2 + 7i$, calculate:

5 $|a + b|$

6 $|a - b|$

7 $|2a + 3b - 5c|$

Calculate the argument, $\arg(z)$, of:

8 $z = 1 + \sqrt{3}\,i$

9 $z = 2 - 2i$

10 $z = -4 + \sqrt{48}\,i$

11 $z = 6 + 6i$

For the complex number $z = 3\sqrt{3} - 3i$, calculate:

12 $|z|$

13 $\arg(z)$.

Solutions

A 13

D 6

E $-45°$ or $\frac{-\pi}{4}$

G $\sqrt{26}$

H $2\sqrt{13}$

I $\sqrt{82}$

L $2\sqrt{10}$

N 5

O $120°$ or $\frac{2\pi}{3}$

R $60°$ or $\frac{\pi}{3}$

S $-30°$ or $\frac{-\pi}{6}$

W $45°$ or $\frac{\pi}{4}$

Y $\sqrt{1306}$